

Leo Fukunaga

5411 Littlebow Road, Rancho Palos Verdes, CA 90275

(424)-383-2075 leofukunaga912@ucla.edu

Education

University of California Los Angeles

- B.S. in **Mathematics of Computation**, expected **June 2029**
- Relevant Coursework: COM SCI 31(**Intro to programming in C++**), COM SCI 32(**Data structures and algorithms**), MATH 33A(Linear Algebra), Math 61(Discrete math) PIC 16A(Programming in Python), MATH 33B(Differential Equations), PIC 16B(Advanced programming in Python)

Los Angeles Harbor College (dual enrollment during high school 2023-2025)

- **A.A. in Liberal Arts and Sciences: Mathematics and Natural Sciences**, February 2025 (Summa Cum Laude)
- Relevant Coursework: **Problem Solving and Program Design Using C++**, **Elementary Statistics with Probability**, **Multi-Variable Calculus**

Palos Verdes Peninsula High School, graduated June 2025

Technical Skills

- **Python, C++, Java, JavaScript, HTML, CSS, SQL, Flask**
- **Git**

Technical Projects ([Portfolio](#))

RMP Professor rating prediction model ([Github Repository](#))

- Built an end-to-end data pipeline to collect, clean, and store **RateMyProfessors data (UC campuses)** using Python, including api usage, preprocessing, and a normalized **SQLite database** for efficient querying.
- Developed a machine learning pipeline using **TF-IDF and NMF-based text features with structured metadata** to predict professor clarity and difficulty ratings, implementing grouped cross-validation and model selection.

League of Legends Match History Dashboard Website ([Github Repository](#))

- Simplified version of [op.gg](#), a League of Legends match history dashboard website

- **Interactive dashboard** where users enter their username to display recent opponents and match history, user rank/level/icon, and specific match info(Kills/Deaths/Assist ratio, Items bought, Character selection, etc...)
- **Python** backend, **HTML**, **CSS** frontend
- Data **obtained and updated live** from Riot Games using [Riot api](#) and [Datadragon](#)

Comparative Analysis Research Project on League of Legends Match Outcomes

- Compared match outcome predictability across varying skill levels to evaluate game balance across skill levels
- Obtained data from Riot Games using [Riot api](#)
- Used **Python** and **Pandas library** to clean and sort **data for over 65,000 games** spanning various skill levels
- Applied various machine learning models from the **scikit-learn library** including **Random Forest**, **Decision Tree**, and **Multi-layer perceptron classifiers**.

Research Experience

Biomedical Research Assistant at The Lundquist Institute, Harbor-UCLA Medical Center, Infectious Disease Lab (June 2024-December 2025)

- Data analysis and interpretation of patient reports for clinical outcome study comparing the efficacy of Trimethoprim Sulfamethoxazole and Doxycycline on uncomplicated skin and skin structure infections
- Created tables and figures for publication manuscript
- Co-author on paper: Miller LG, Phan Tran D, Liu H, Boyle M, Flores E, Pathmarajah K, Moran G, Munekata M, Zhou L, Williams P, Han B, **Fukunaga L**, Fritz S. A Randomized Trial of Doxycycline vs. Trimethoprim Sulfamethoxazole (TMP-SMX) for Uncomplicated Skin and Skin Structure Infections (SSSIs). Accepted to IDWeek 2025, October 19-22 2025, Atlanta GA.
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12793100/>